

ZHYTOMYR, Ukraine

WMO: 333250

Lat: 50.2806N

Lon: 28.6592E

Elev: 735

StdP: 14.31

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.5	5.2	-5.1	4.3	0.4	0.0	5.6	6.4	22.5	34.6	19.2	32.5	3.5	310	0.500

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	86.6	68.8	83.6	67.3	80.9	66.2	71.1	82.6	69.5	79.9	67.9	77.9	5.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	103.0	76.4	65.7	97.4	74.9	64.2	92.3	73.1	35.4	83.2	34.0	79.7	32.7	77.9	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
18.4	15.9	13.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-7.0	90.8	7.0	3.6	-12.0	93.3	-16.1	95.4	-20.1	97.4	-25.2	100.0		
			-7.4	73.7	6.9	2.0	-12.4	75.1	-16.4	76.2	-20.3	77.4	-25.3	78.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	25.2	27.4	35.9	49.0	58.8	64.7	68.1	66.5	57.3	46.6	37.4	28.6	
	DBStd	17.25	10.25	10.28	8.30	7.55	7.45	6.09	5.36	5.93	6.98	7.82	8.47	9.87	
	HDD50	3186	768	634	439	106	14	0	0	0	16	159	385	665	
	HDD65	6825	1233	1054	901	482	219	81	30	51	245	570	829	1130	
	CDD50	2175	0	0	3	76	286	442	560	512	236	54	6	0	
	CDD65	337	0	0	0	2	26	72	125	98	14	0	0	0	
	CDH74	3074	0	0	0	19	265	627	1063	950	148	2	0	0	
	CDH80	781	0	0	0	3	43	122	289	295	29	0	0	0	
Wind	WSAvg	5.5	6.4	6.7	6.5	5.8	5.2	4.8	4.4	4.2	4.7	5.4	6.1	5.9	
Precipitation	PrecAvg	23.9	1.3	1.1	1.2	1.7	2.3	3.0	3.5	3.0	2.0	1.4	1.8	1.5	
	PrecMax	34.6	3.7	3.2	2.7	3.8	4.5	6.2	8.1	6.8	7.3	4.2	4.9	3.0	
	PrecMin	12.7	0.4	0.2	0.0	0.4	0.0	0.6	0.6	0.7	0.2	0.1	0.4	0.1	
	PrecStd	4.3	0.8	0.7	0.8	0.9	1.1	1.4	2.0	1.7	1.5	1.0	1.0	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.2	50.8	61.5	76.2	84.6	86.5	90.5	91.7	83.7	72.7	60.6	50.8	
		MCWB	41.9	44.5	49.6	59.0	64.9	68.5	70.7	70.4	64.7	59.6	53.5	47.4	
	2%	DB	42.5	45.1	56.4	70.1	80.3	83.5	86.5	86.9	78.0	67.1	55.0	45.9	
		MCWB	39.4	40.7	46.5	55.4	63.2	66.9	69.6	68.6	62.6	57.7	50.6	43.4	
	5%	DB	39.4	42.0	51.9	65.9	76.8	80.9	83.2	83.0	73.9	62.0	51.9	42.9	
		MCWB	37.2	38.9	44.1	52.9	62.0	66.3	68.6	66.9	61.1	55.2	48.7	41.0	
	10%	DB	36.8	39.0	47.7	62.2	73.0	77.7	80.1	79.3	69.6	58.1	48.6	39.9	
		MCWB	35.1	36.4	41.9	51.0	60.1	64.7	67.3	65.6	59.5	52.8	46.2	38.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.5	45.7	50.6	60.3	68.0	71.7	73.1	73.2	67.2	62.2	54.5	47.9
			MCDB	45.2	50.1	59.4	74.9	81.3	81.5	85.1	85.7	79.4	69.9	58.7	50.1
2%		WB	39.9	41.6	47.6	56.9	65.0	69.5	71.4	70.6	64.4	58.8	51.3	43.6	
		MCDB	41.9	44.5	54.3	66.6	76.2	79.6	83.4	82.3	74.4	64.9	54.5	45.4	
5%		WB	37.4	39.0	44.9	54.5	63.2	67.7	69.9	68.7	62.3	55.9	48.9	41.2	
		MCDB	39.3	41.6	50.5	63.4	74.0	77.9	80.1	79.5	71.3	61.2	51.4	42.8	
10%		WB	35.2	36.7	42.2	52.3	61.3	65.9	68.3	66.9	60.3	53.2	46.5	38.2	
		MCDB	36.5	38.8	47.2	60.2	70.7	75.2	77.7	76.7	68.0	57.3	48.4	39.8	
Mean Daily Temperature Range	5% DB	MDBR	7.8	9.5	12.3	16.7	18.8	17.9	18.0	19.1	16.6	14.1	8.6	7.6	
		MCDBR	7.9	11.7	19.2	22.8	23.6	22.6	23.0	24.9	23.8	20.1	13.7	9.2	
		MCWBR	6.5	8.7	11.3	11.0	10.1	9.5	9.1	9.6	10.8	11.5	9.5	7.5	
	5% WB	MCDBR	7.4	10.8	17.0	20.1	21.4	19.8	20.2	21.7	20.8	18.3	11.7	8.4	
		MCWBR	6.5	8.5	10.9	10.6	10.1	9.7	9.1	9.5	10.5	11.1	8.7	7.4	
Clear-Sky Solar Irradiance	taub	0.309	0.339	0.376	0.431	0.405	0.401	0.430	0.431	0.397	0.360	0.341	0.307		
	taud	2.207	2.161	2.187	2.124	2.275	2.306	2.235	2.227	2.280	2.355	2.302	2.259		
	Ebn at Noon	223	246	259	257	270	272	261	254	250	238	210	205		
	Edh at Noon	24	32	37	45	40	39	42	40	34	27	22	20		
All-Sky Solar Radiation	RadAvg	275	529	934	1376	1773	1879	1793	1576	1083	637	261	189		
	RadStd	49	102	149	145	151	137	124	89	167	100	30	45		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.76	N/A	N/A	+1.73	+1.42	+1.10	-376	-533	+293	+122
(47)	Regional (10 neighbors)	+1.30	N/A	N/A	+1.73	+0.99	+0.90	-243	-355	+214	+90

Nomenclature: See separate page